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Unit – 3 : Session – 4 : SLO – 1 | Stack Applications — Reversing a List and Balancing Symbols

■ I. Reversing a List Using a Stack

- Step 1: List A = [10, 20, 30, 40, 50] (5 elements).
- Step 2: Traverse A from the first element and push each onto Stack S:
- Push 10 -> S = [10]; Push 20 -> [10,20]; Push 30 -> [10,20,30];
- Push 40 -> [10,20,30,40]; Push 50 -> [10,20,30,40,50]. TOP = 50.
- Step 3: Pop from TOP one by one into array B:
- Pop 50 -> B[0]=50; Pop 40 -> B[1]=40; Pop 30 -> B[2]=30;
- Pop 20 -> B[3]=20; Pop 10 -> B[4]=10.
- Result: B = [50, 40, 30, 20, 10] — the reversed list.

■ II. Balanced Parenthesis Check: { P + (Q - R) }

- flag = 1. Scan left to right:
- '{' -> push. Stack: {
- '(' -> push. Stack: { (
- ')' -> pop '(' -> matches '('. Stack: {
- '}' -> pop '{' -> matches '{'. Stack: EMPTY.
- End of expression: stack empty and flag = 1 -> VALID (balanced).

■ III. Balanced Parenthesis Check: { D * (S + A) }

- flag = 1. Scan left to right:
- '{' -> push. Stack: {
- '(' -> push. Stack: { (
- ')' -> pop '(' -> matches. Stack: {
- '}' -> pop '{' -> does NOT match '}' -> flag = 0.
- End of expression: flag = 0 -> INVALID (unbalanced).